Thuan Hieu Tran

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Education

University of Chicago — Chicago, IL
B.S. in Computer Science & Economics | Full-Ride Award — *Expected June 2030*

Experience

Software Engineer & Project Lead — Feb. – Aug. 2026
IterLight — *Florida, USA (Remote)*

- Revived a previously failed initiative as project lead: built a custom harness for a human-in-the-loop AI pipeline that auto-generates TikTok videos from the company's proven viral format
- Scaled the pipeline from 1 to 5 TikTok channels with a scheduled multi-account posting system; the top channel passed 3M views at roughly 90% lower cost than building an in-house production team
- Solo-built a RAG sales and support pipeline for Facebook Messenger: architected a 138-page linked-wiki knowledge base, designed a human-review workflow and unlocked personal-account chats beyond the official API via Matrix
- Ran continuous pilot-and-refine cycles with the sales team, doubling reply speed and raising the conversion rate 1.5x; the pipeline replaced outsourced sales agents, with API token costs 60% below projections

Research Assistant — 2025
AI4LIFE Institute, Hanoi University of Science and Technology — *Hanoi, Vietnam*

- Built the data preprocessing pipeline for a spatial gene expression model on H&E histology slides, turning 36 whole-slide images from 8 patients into 9,612 spot-level patches across 785 genes
- Fine-tuned the UNI pathology foundation model with LoRA to predict gene expression from histology images; final model reached 3.5x the baseline Pearson correlation (PCC)

Algorithm Optimization Intern — June – Sep. 2024
AimeSoft — *Hanoi, Vietnam*

- Optimized train timetables for a Japanese rail operator client via greedy-initialized simulated annealing
- Reformulated the objective function to incorporate previously unused sensor data; validated schedules in a discrete-event simulation, cutting worst-case passenger wait time by up to 17%

Honors & Awards

Silver Medal, International Artificial Intelligence Olympiad (Ljubljana, Slovenia; UNESCO/AWS)
Third Prize, ICPC Asia Regional; **First Prize (7th of 200 teams)**, ICPC Vietnam (High School Division)
National Champion (1st of 2,200), Vietnam National Young Informatics Contest
Champion, Vietnam AI Championship; **Second Prize**, Vietnam Olympiad in AI
National Third Prize (top 8%), Vietnam Olympiad in Informatics (VOI); **Codeforces Master** (rating 2104)

Research & Publications

Sign Language Recognition: A Large-scale Multi-view Dataset and Comprehensive Evaluation
Co-Author — *WACV 2025*

- Co-developed Multi-VSL, a pioneering multi-view Vietnamese Sign Language dataset (84K+ videos, 1,000 glosses, 30 signers) that became a foundation for later VSL research and products in Vietnam; 26 citations
- Set up the 3-camera recording studio and built the annotation platform; cross-tested I3D, Swin, MViTv2 and VTN-PF across views, showing multi-view fusion lifts top-1 accuracy by up to 19.75 pp

Projects

LittleHands | Co-Creator & Lead Developer — June 2024 – Present
A smartphone app that translates Vietnamese Sign Language to and from text for deaf and hard-of-hearing users

- Trained Video Transformer Networks on our 84K-video Multi-VSL dataset for sentence-level sign recognition, powering the app's sign-to-text translation
- Built the FastAPI + PostgreSQL backend and C# client for two-way sign, text and animation translation

Bottleneck-Migration Alpha — July 2026

- Built a market-neutral long/short sector-rotation strategy on the AI hardware supply chain, with price-based regime detection, beta hedging and volatility targeting; backtested Sharpe 1.30 net of costs

Technical Skills

Languages & ML: Python, C/C++, Java, PyTorch, CUDA, NumPy, pandas, Hugging Face (PEFT/LoRA), Unsloth
Computer Vision: OpenCV, torchvision, video recognition models (I3D, Swin, MViTv2, VTN)
Infrastructure: Google Cloud, Docker, Matrix (Synapse), Linux/Bash, Git